

JavaScript Loop and Functions

Tasks

In this task you are asked to construct an **Ice Cream Ratings List** that is dynamically extended when valid input from the user is received through a form.

You are provided with a starting web page, that contains a form (two fields) and a list that is already populated with one rating. The sample starting page should look like this:

Ice Cream Ratings Page

localhost:3000

Incognito

World of Ice Cream

Provide your rating on your recent Ice Cream experience

Javascript Loops and Functions

A web pages can be a great way of getting feedback from users. Here, we are asking the user for their name and a rating on a recent *Ice Cream tasting experience* they have had. After inputting their name, the use gives how many *stars* out of **5** they would like to give. This is then recorded and added to a list of previous results.

NOTE: The use should only be able to input a number between 0 and 5, with an error being displayed if it is outside these bounds.

Input Fields

Name:

Star rating (between 0 and 5):

Ratings list last updated on 23/03/2024 at 14:26:17:

- [23/03/2024 14:26:17] **Mick Hobbs** gave a rating of ★★☆☆☆

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Initial page view

The form accepts a `Name` as text and a value representing the `Number of Stars` the user rates the ice cream. The information is then added to the list, with three parts:

1. The date/time of the input
2. The name of the user giving the rating
3. The star rating (number of stars) given as a sequence of filled & un-filled star characters.

The following screenshots show the result of three additional user responses. The first is *'Sam Smith' with a rating of 4 out of 5 stars*:

and added to a list of previous results.

NOTE: The use should only be able to input a number between 0 and 5, with an error being displayed if it is outside these bounds.

Input Fields

Name:

Star rating (between 0 and 5):

[Submit a Ice Cream Rating](#)

Ratings list last updated on 23/03/2024 at 15:32:08:

- [23/03/2024 14:26:17] **Mick Hobbs** gave a rating of ★★☆☆☆
- [23/03/2024 15:32:08] **Sam Smith** gave a rating of ★★★★★

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Input of *'Sam Smith' with a rating of 4 out of 5 stars*

The next is *'Eric banner' with a rating of 0 out of 5 stars*:

NOTE: The use should only be able to input a number between 0 and 5, with an error being displayed if it is outside these bounds.

Input Fields

Name:

Star rating (between 0 and 5):

[Submit a Ice Cream Rating](#)

Ratings list last updated on 23/03/2024 at 15:32:32:

- [23/03/2024 14:26:17] **Mick Hobbs** gave a rating of ★★☆☆☆
- [23/03/2024 15:32:08] **Sam Smith** gave a rating of ★★★★★
- [23/03/2024 15:32:32] **Eric Banner** gave a rating of ☆☆☆☆☆

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Input of *'Eric Banner' with a rating of 0 out of 5 stars*

And the final screenshot shows *'Lara Croft' with a rating of 5 out of 5 stars*:

NOTE: The user should only be able to input a number between 0 and 5, with an error being displayed if it is outside these bounds.

Input Fields

Name:

Star rating (between 0 and 5):

Submit a Ice Cream Rating

Ratings list last updated on 23/03/2024 at 15:34:11:

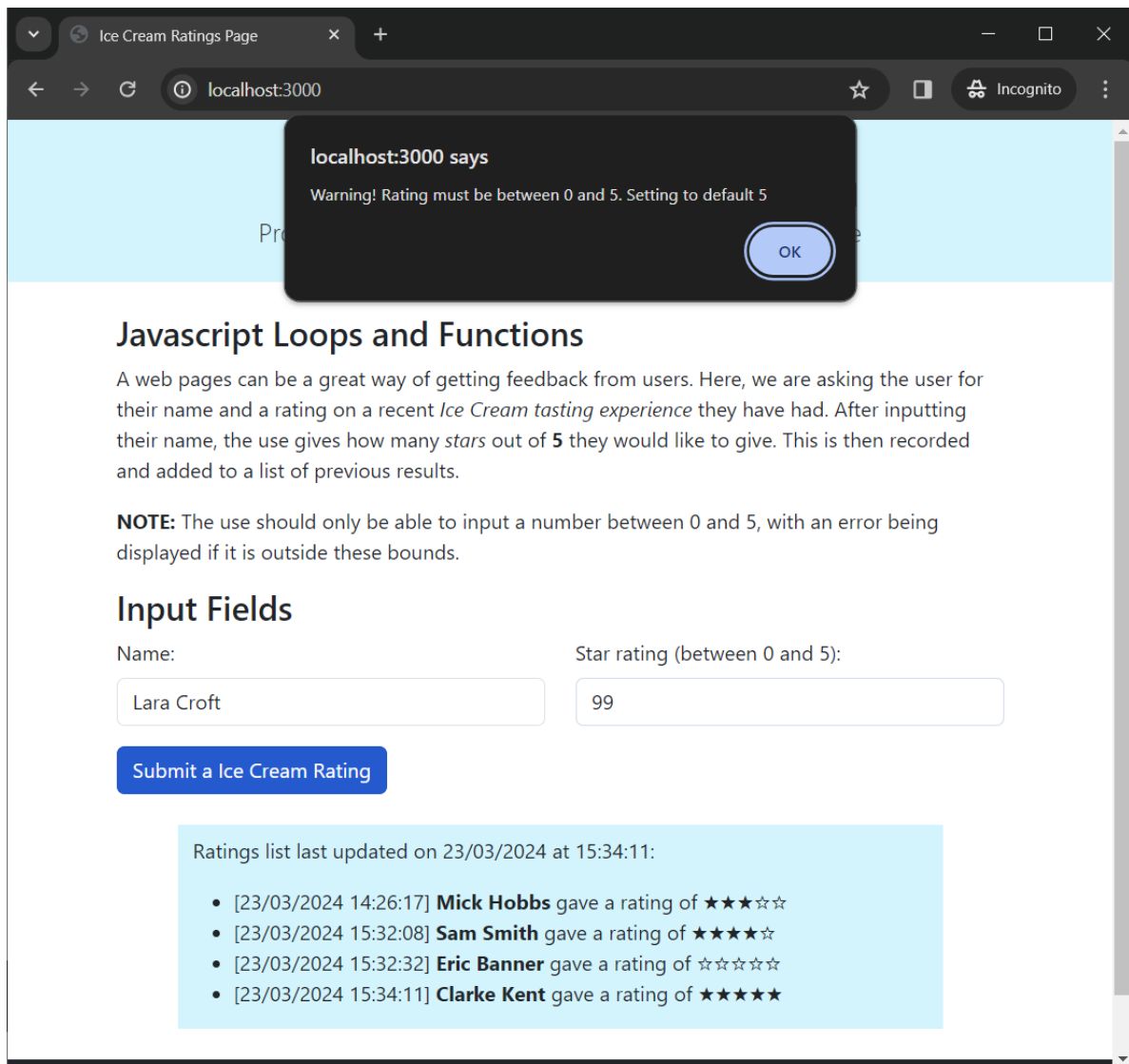
- [23/03/2024 14:26:17] **Mick Hobbs** gave a rating of ★★☆☆☆
- [23/03/2024 15:32:08] **Sam Smith** gave a rating of ★★★★★
- [23/03/2024 15:32:32] **Eric Banner** gave a rating of ☆☆☆☆☆
- [23/03/2024 15:34:11] **Clarke Kent** gave a rating of ★★★★★

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Input of 'Clarke Kent' with a rating of 5 out of 5 stars

In completing this task, the code used to dynamically add ratings to the list should be allocated within a javascript function. This function is called when the user clicks the `Submit an Ice Cream Rating` button.

In adding to the list, the javascript code should take and process the input fields from the form (`nameInput` and `ratingInput`). The *rating* input must be checked to be within the 0..5 range and if it is outside this range an error message provided. In the error case the rating can be set to a maximum of 5. A sample of the error prompt provided could be:

Error message on invalid *rating* value

Hints

In completing this task consider the following points:

- Build your code in stages, i.e., get and check input from the user; find the fields within the HTML document that need to be updated (e.g., `last-updated-date` `last-updated-time` and the `ratingsList`)
- The `input` fields of the form can be extracted using DOM commands such as
 - `let name = document.getElementById("nameInput").value;`
 - `let rating = document.getElementById("ratingInput").value;`
- The rating value inputted by the user should be checked to see it is **invalid** (i.e., `< 0` or `> 5`) and if it is an error message should be provided (using `window.alert()`) along with setting it to a default rating of **5**.

- Before you can add a new *list element* the text it contains should be created. Parts of this text use the *Date()* methods:

```
let now = new Date();
let date = now.toLocaleDateString();
let time = now.toLocaleTimeString();
...
```

- You can use a `for()` loop to add the number of stars to the end of this text. The HTML symbols (`★` and `☆` can be used for the ★ and ☆ symbols). The filled stars can be added using the `for( let i=0; i<rating; i++) {..}` loop, while the unfilled stars with `for( let i=0; i<(5-rating); i++){..}`.
- Online references include:
 - (https://www.w3schools.com/js/js_dates.asp)
 - (https://www.w3schools.com/jsref/dom_obj_table.asp)
 - (https://www.w3schools.com/charsets/ref_utf_symbols.asp)

What will you submit?

You should submit:

- Screen-shot of the web page showing the output for 5 added valid ratings (total of 6 names/rating in list).
- Screen-shot of the page when an 'invalid' rating is entered (i.e., a rating of `-1` or `99`) and the prompt warning window is presented.
- The webpage's `HTML` file.
- The file with `JavaScript` code used to build the table.